

Lab 6 — Email Header & PGP/GPG Analysis

Thunderbird · Gpg4win · Command-line GPG verification

Objective

Investigate email headers and PGP-signed messages using Thunderbird and Gpg4win. Analyze email headers to trace sender IP address and relay servers. Learn to verify authenticity and integrity of encrypted and digitally signed messages using GPG tools.

Screenshot 1 — Thunderbird: Raw Email Headers

Open the suspicious email → **More** ■ **View** ■ **Message Source** (Ctrl+U). The raw headers reveal the full delivery path, authentication results, and content type.

Header Field	Value
Return-Path	<alice@example.com>
Received (hop 2)	from mail.example.com (203.0.113.42) by mx.recipient.org Tue, 21 Apr 2026 14:32:11 +0530
Received (hop 1)	from [192.168.1.5] (smtp.example.com [198.51.100.10]) by mail.example.com id a3sd7fg
MIME-Version	1.0
From	Alice Smith <alice@example.com>
To	Bob Jones <bob@recipient.org>
Subject	[Signed] Meeting Notes – Q2 Review
Date	Tue, 21 Apr 2026 14:31:50 +0530
Message-ID	<CABc3fGh789xyz@mail.example.com>
Auth-Results spf	pass smtp.mailfrom=alice@example.com
Auth-Results dkim	pass header.d=example.com
Auth-Results dmarc	pass (policy=reject) header.from=example.com
Content-Type	multipart/signed; micalg=pgp-sha256; protocol="application/pgp-signature"

Screenshot 2 — Sender IP & Relay Server Trace

Parse the **Received** headers from bottom to top to reconstruct the email's delivery path. The bottom-most Received header is the first hop (closest to the sender).

Relay Hop Trace

Hop	Host / IP	Timestamp	Note
1 (Origin)	192.168.1.5	14:31:50 +0530	Sender's LAN IP (private)
2 (Relay)	smtp.example.com [198.51.100.10]	14:31:58 +0530	Outgoing SMTP relay
3 (Relay)	mail.example.com [203.0.113.42]	14:32:05 +0530	Domain mail server

4 (Dest MX)	mx.recipient.org	14:32:11 +0530	Final delivery — PASS
-------------	------------------	----------------	-----------------------

SPF / DKIM / DMARC Authentication Results

Check	Result	Detail
SPF	PASS	smtp.mailfrom=alice@example.com — authorized sender IP
DKIM	PASS	header.d=example.com — signature cryptographically valid
DMARC	PASS	policy=reject — domain owner enforces strict alignment

Screenshot 3 — GPG CLI Signature Verification

Use **gpg --verify** to check a PGP-signed message. A **Good signature** confirms the message was not altered and was signed by the claimed key. A **BAD signature** means the content was tampered after signing.

Case A — Verifying a legitimate signed message:

```
$ gpg --verify email_body.asc email_body.txt
gpg: Signature made Tue 21 Apr 2026 14:31:50 IST
gpg: using RSA key 4A3B7F89C12D56EF0123456789ABCDEF01234567
gpg: checking the trustdb
gpg: marginals needed: 3 completes needed: 1 trust model: pgp
gpg: Good signature from "Alice Smith <alice@example.com>" [full]
Primary key fingerprint: 4A3B 7F89 C12D 56EF 0123 4567 89AB CDEF 0123 4567
```

Case B — Verifying a tampered / forged message:

```
$ gpg --verify tampered_email.asc tampered_body.txt
gpg: Signature made Tue 21 Apr 2026 14:31:50 IST
gpg: using RSA key 4A3B7F89C12D56EF0123456789ABCDEF01234567
gpg: BAD signature from "Alice Smith <alice@example.com>" [full]
```

Key Listing Command

```
$ gpg --list-keys alice@example.com
pub rsa4096 2024-01-15 [SC] [expires: 2027-01-14]
4A3B7F89C12D56EF0123456789ABCDEF01234567
uid [full] Alice Smith <alice@example.com>
sub rsa4096 2024-01-15 [E] [expires: 2027-01-14]
```

Screenshot 4 — Kleopatra (Gpg4win GUI): Certificate Manager

Kleopatra provides a graphical interface for managing GPG keys and verifying signed files. Use **File ■ Decrypt/Verify Files** and select the .asc signature file.

Certificate Manager — Imported Keys

Name / Email	Key ID	Validity	Expires
Alice Smith <alice@example.com>	89ABCDEF01234567	Full	2027-01-14
Bob Jones <bob@recipient.org>	AABBC1122334455	Unknown	2026-12-31

Decrypt/Verify Result — email_body.asc

```
✓ Verification successful
Signed on: Tue, 21 Apr 2026 14:31:50 +0530
Signed by: Alice Smith <alice@example.com>
Key ID: 89ABCDEF01234567
Fingerprint: 4A3B 7F89 C12D 56EF 0123 4567 89AB CDEF 0123 4567
GOOD SIGNATURE – message integrity intact
```

Lab Summary

Task	Tool Used	Key Finding
View raw email headers	Thunderbird (Ctrl+U)	Full Received chain, Message-ID, Auth results
Trace sender IP & relays	Manual header parsing	Origin: 192.168.1.5 → 2 relays → mx.recipient.org
Check SPF/DKIM/DMARC	Auth-Results header	All three checks: PASS
Verify good signature (CLI)	gpg --verify	Good signature — integrity confirmed
Detect tampered message (CLI)	gpg --verify	BAD signature — content modified
GUI certificate management	Kleopatra (Gpg4win)	Full-trust key verified; signature valid